

Exam-like Problems Project 02

Q1 $a0=1, a1=2, a2=3$

$\text{add } a0, a0, a0 \quad \# a0=2$

$\text{add } a1, a1, a2 \quad \# a1=5$

$\text{mul } a2, a1, a0 \quad \# a2=10$

$a2$ is 10

Q2 $a0=0, a1=3$

$\text{addi } a0, a0, 1 \quad \# a0=1$

; boo

foo:

$\text{addi } a0, a0, a1 \quad \# a0=3+3=6$

; goo

boo:

$\text{addi } a0, a0, 2 \quad \# a0=3$

$\text{beq } a0, a1, foo \quad \# \text{jump to foo}$

goo:

$\text{addi } a0, a0, 1 \quad a0=6+1=7$

$a0$ is 7

Q3 $a0=0, a1=2, a2=0$

$\text{li } a3, 0$

loop:

$\text{beq } a1, a2, loopend$

$\text{add } a3, a3, a1 \quad \# a3=2 \text{ then } a3=3$

$\text{addi } a1, a1, -1 \quad \# a1=1 \text{ then } a1=0$

; loop

loopend:

$\text{mv } a0, a3 \quad \# a0=3 \text{ because } a3=3$

$a0$ is 3 and there was 11 instructions I wrote the instruction # on the right in-line with the code.

Instruction count

1

2

3

4

5

6

7

8

9

10

11